The Climate Standoff

Introduction:

Threshold public goods models explain coordination failure through two interlocking mechanisms: collective benefits depend on reaching a critical participation mass, and multiple equilibria make that mass difficult to attain. The literature has produced elegant characterizations of why cooperation stalls. It has been less successful at explaining why it sometimes accelerates.

The most consequential shifts in recent climate policy were not preceded by formal treaties. The European Union's Carbon Border Adjustment Mechanism did not emerge from multilateral negotiation. It became politically viable only after domestic decarbonization created the incentive to export climate standards through trade policy. The United States Inflation Reduction Act reshaped global clean-energy investment through large-scale subsidies, triggering competitive policy responses across Europe and Asia within eighteen months. In both cases, the incentive to participate was generated endogenously: early movers altered the payoff matrix for everyone else, and the institutional architecture followed strategic logic rather than producing it.

The strategic environment is best understood as a multilateral game of chicken among highly asymmetric players. The blocs are not driving identical cars toward the same obstacle. A small number of players are systemically important enough that their failure to act carries disproportionate global consequences, placing them under asymmetric pressure to move first. Others face weaker incentives to adjust early, because short political horizons and high transition costs raise the price of unilateral action. Still others sit closer to the cliff: climate damages arrive faster, hit more vulnerable sectors, and leave less room to wait. The coordination problem is therefore shaped not only by whether blocs eventually cooperate, but by who can afford to delay, who bears the greatest cost of collective failure, and whose move most strongly updates the incentives of every other player.

Standard treaty-based models abstract from these dynamics. By treating enforcement pressure, cost trajectories, and threshold progress as exogenous features of the environment rather than outcomes of strategic interaction, they struggle to account for the timing, escalation, and asymmetry that define modern climate geopolitics. A framework that takes those variables seriously can explain why cooperation deepens even when formal agreements stagnate.

This paper develops a finite-horizon dynamic Markov game to analyze these interactions. Heterogeneous blocs face binary adoption decisions in each period. Transition costs decline endogenously through learning-by-doing; political pressure rises with each new adopter; collective benefits are realized only once a weighted participation threshold is crossed. The model delivers three results. First, anchor players whose adoption most strongly shifts the threshold calculus for others can trigger tipping dynamics that transform a dangerous game of delay into a self-reinforcing race toward cooperation. Second, asymmetric patience is the primary lever governing whether tipping occurs before or after irreversible damage accumulates. Third, the policy implications differ sharply depending on whether laggards are strategic free-riders or capacity-constrained actors facing a genuinely different cost schedule.

Equilibrium Concepts : Nash to Quantal Response

The dominant solution concepts in the climate cooperation literature share a common assumption: players best-respond exactly to the strategies of others. Barrett (1994) uses stage-game Nash equilibrium and renegotiation-proof repeated-game equilibrium to characterize self-enforcing international environmental agreements. Harstad (2016) selects Markov perfect equilibrium on the grounds that strategies depending only on payoff-relevant state variables are both analytically tractable and behaviorally plausible. The assumption of exact best response is not incidental to these frameworks. It is load-bearing.

Within the class of models Barrett inaugurates, this assumption generates a well-documented fragility. Participation decisions are discrete, and stable coalitions are determined by internal and external stability conditions. Small parameter changes therefore induce large, discontinuous shifts in equilibrium coalition size. The framework moves abruptly between near-full cooperation and near-complete defection, which limits its usefulness precisely where the empirical record is most interesting: the margin where marginal changes in incentives produce cascading adoption rather than stasis.

Quantal response equilibrium resolves this by replacing exact best response with probabilistic choice. Players choose better options more often, but not always. Choice probabilities increase in expected payoffs, governed by the rationality parameter lambda. As lambda rises, behavior approaches the deterministic best responses of Nash equilibrium. As lambda falls, choices converge toward randomness. A marginal increase in adoption costs therefore reduces the probability of adoption incrementally rather than flipping participation off entirely.

The calibrated value of lambda carries a direct substantive interpretation here. It captures the noise inherent in international negotiations where governments are constrained by domestic win sets, bureaucratic friction, and electoral uncertainty. Stochastic choice is not a technical device to smooth the equilibrium correspondence. It reflects a world in which sovereign states cannot act as perfectly rational best-responders even when the strategic logic of cooperation is clear. By situating the model in this two-level context, QRE provides a more accurate representation of how governments navigate the tension between international commitments and domestic survival.

Equilibrium is found at the fixed point where beliefs equal behavior: each bloc's strategy is optimal given what it expects others to do, and what it expects others to do is exactly what they do. The fixed-point algorithm converges to a consistent solution across initializations, confirming that the equilibrium is not an artifact of the starting condition.

Resolving the fragility problem is necessary but not sufficient. QRE ensures that smooth changes in incentives produce smooth changes in behavior. It does not by itself explain when those smooth changes accumulate into the self-reinforcing adoption cascades that define the empirical transitions this paper aims to characterize. That requires a theory of tipping.

Strategic tipping and diffusion dynamics

Quantal response equilibrium explains how equilibrium behavior responds smoothly to shifting incentives. It does not explain when smooth responses accumulate into cascades. A separate body of work addresses this directly.

Farmer et al. (2019) introduce sensitive intervention points in socioeconomic systems, arguing that small, well-targeted interventions can exploit amplification mechanisms to trigger cascading low-carbon transitions. Otto et al. (2020) identify specific social subsystems where positive feedback could rapidly compress emissions trajectories. Sharpe and Lenton (2021) extend this into a framework of upward-scaling tipping cascades, in which activating one socio-technical tipping point raises the probability of triggering the next. Way et al. (2022) show that endogenous cost declines through fast learning curves for renewables can shift mitigation pathways more sharply than most integrated assessment models assume.

These contributions capture tipping dynamics observed in real energy transitions. They share a structural limitation. Adoption decisions are embedded in aggregate cost functions or representative-agent frameworks rather than arising from strategic interaction between heterogeneous countries. This matters because the mechanism is unresolved. A cascade driven by falling costs has different policy implications than one driven by shifting expectations or peer pressure from anchor adopters. Without a game-theoretic foundation, the models cannot separate these channels, and cannot explain why some countries adopt early while structurally similar ones delay.

A second problem compounds the first. Tipping behavior in formally specified models is often sensitive to the chosen parameter values, which makes it difficult to distinguish structurally robust dynamics from calibration artifacts. A finding that holds only at a particular \lambda
Or cost trajectory is not a finding about tipping. It is a finding about one point in the parameter space.

Taken together, these gaps define the space this paper occupies. The literature needs a framework that endogenizes tipping through asymmetric strategic interaction, and a disciplined method for establishing which tipping dynamics survive joint parameter perturbation. The next section turns to the third strand of relevant work: the formal modeling of international environmental agreements and what the coalition stability literature contributes, and fails to contribute, to both problems.

Gaps and Contribution

Two gaps run through the literature reviewed above. The first is structural. No existing framework models declining costs, endogenous political pressure, and asymmetric influence as joint determinants of cooperation. Agreement-based models treat enforcement as exogenous. Diffusion models embed adoption in aggregate cost functions rather than strategic interaction. Neither can explain how early adoption by anchor players reshapes the incentives of strategic holdouts over time, because neither gives those players a strategic role.

The second gap is methodological. Game-theoretic models of climate cooperation are known to be sensitive to parameterisation, yet few studies distinguish the cross-bloc structure that is pinned down by observable data from the latent cardinal strength of the underlying payoff channels. Most computational approaches calibrate to a single benchmark or rely on ad hoc sensitivity checks. This makes it difficult to tell whether a finding reflects a structural property of the game or a feature of one point in the parameter space.

This paper addresses both gaps by embedding these mechanisms in a finite-horizon dynamic Markov game with heterogeneous blocs and irreversible adoption. Behaviour is characterised using quantal response equilibrium, which allows smooth comparative statics and realistic strategic delay without the brittleness of Nash-based coalition models.

The methodological contribution lies in separating calibration into two layers. The ordinal layer is identified directly from macroeconomic and emissions data, preserving the empirical asymmetries across blocs without imposing arbitrary structure. The cardinal layer is then recovered using a simulated method of moments, matching observable features of the contemporary climate transition: EU leadership, strategic free-riding by major emerging economies, and the delayed onset of self-reinforcing adoption. Together these produce a single fitted strategic baseline that starts from a credible representation of the 2025 environment.

From that baseline, the analysis identifies which policy levers most reliably advance earlier coordination. Local sweeps establish which channels matter in isolation. Global sensitivity analysis then stress-tests the rankings by jointly perturbing the full parameter space, separating levers that are locally powerful from those that remain influential once the wider strategic environment moves. The distinction matters for policy: a lever that works only near the baseline calibration is not a robust prescription.

Model Setup / Math (see overleaf doc)

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Calibration

Implementing the model requires assigning numerical values to all parameters. Some can be disciplined directly by observable data. Others cannot. The calibration strategy addresses this by separating the problem into two layers, each with a distinct identification logic.

The first layer fixes ordinal structure: the relative positioning of blocs within each parameter channel. Which bloc faces the highest transition costs, which is most exposed to climate damages, which exerts the greatest systemic influence over others these cross-bloc rankings can be inferred from data. World Bank development indicators, national emissions inventories, and bilateral trade statistics provide consistent evidence on the relative positions of the United States, European Union, China, and the Rest of the World. This layer is treated as empirically identified and held fixed throughout the analysis. Perturbing it would mean contradicting the observable record.

The second layer recovers cardinal scaling: the absolute magnitude of each payoff channel and its weight in shaping strategic behavior. The relative importance governments place on transition costs versus climate damages versus political pressure is not directly observable and cannot be read off macroeconomic data. Rather than treating these weights as free scenario assumptions, I estimate them using a simulated method of moments (SMM).

The moments are chosen to represent the observable strategic fingerprint of the 2025 climate environment. I target the renewable adoption ratios between the EU and its peers, the near-certain probability of EU first-mover behavior, and the absence of near-term global coordination cascades. Each moment captures a distinct feature of the strategic landscape: the cost channel leaves a fingerprint in adoption gaps, leadership probabilities identify the pressure channel, and the absence of cascades pins down the threshold structure. Together they anchor the model to the contemporary environment rather than to a stylised benchmark.

The fitted parameter vector is therefore the one that makes the model reproduce these observable features of the world. Ordinal structure is data-identified. Cardinal scaling is structurally estimated. The sensitivity analysis in the results section is conducted around this fitted baseline, which is what allows the distinction between locally powerful levers and structurally robust ones to carry empirical weight.

Layer 1 : Ordinal Structure

Per Player Params:

Each per-player parameter is constructed from bloc-level data through a structural transformation anchored to the EU as a reference bloc. This preserves empirical ordinal relationships across blocs while a single scaling parameter alpha determines the absolute level.

Transaction Costs: are derived from the carbon intensity of GDP, adjusted for development. Carbon intensity captures the extent of structural change required for decarbonisation: higher-intensity economies must replace more emissions-intensive capital to meet the same mitigation target. Raw intensity cannot be taken at face value for less developed economies, however. Low carbon intensity in those cases often reflects limited industrialisation rather than genuinely clean infrastructure, which would falsely register as low transition costs.

To correct for this, I apply a development-gated interpolation. Each bloc's effective carbon intensity is defined as a weighted average of its observed intensity and a global baseline, where the weight rises with GDP per capita relative to the richest bloc. Developed economies retain their observed advantage. Less developed economies are partially adjusted toward the baseline, preventing the model from treating structural underdevelopment as a decarbonisation head start.

An affordability penalty is then applied on top. The cost of capital for renewable energy projects in developing economies is empirically two to three times higher than in advanced economies, reflecting financing constraints and weaker deployment capacity (Ameli et al., 2021). The affordability elasticity (epsilon = 0.3) governs how steeply effective costs rise as income falls below the richest bloc. The chosen value reproduces cost differentials of that order of magnitude within the model. Together the two adjustments ensure the ordinal cost structure reflects genuine transition burdens rather than artefacts of the raw intensity data.

Climate damages: Climate damages are proxied by the share of agriculture, forestry, and fishing in GDP. This captures structural economic exposure to climate risk: economies with larger primary sectors face greater direct losses from temperature extremes, water stress, and ecosystem disruption. More composite vulnerability indices exist, but agriculture share is preferred for its direct observability and its consistency across blocs in World Bank national accounts data. The proxy is a deliberate choice between precision and comparability, and comparability wins at this level of aggregation.

Political pressure: Political pressure is proxied by trade openness, measured as trade as a share of GDP. A more trade-dependent economy has more to lose from non-cooperation: carbon border adjustments, climate-linked conditionality, and broader diplomatic friction all arrive through trade relationships. Trade openness therefore captures the degree to which a bloc's economic welfare is hostage to the policy choices of its major partners, and how exposed it is to externally imposed costs if it delays.

Influence weights: Influence weights are defined as an equal-weighted combination of emissions share and GDP share. Emissions share captures a bloc's necessity for any effective global agreement. GDP share captures its economic leverage within the international system. Equal weighting reflects the assumption that neither dimension dominates the other.

All parameters are normalised relative to the EU benchmark, so each scaling parameter alpha Directly sets the EU's level for a given channel. The values for other blocs are then determined by their data-implied ratios to the EU. This structure limits the modeller's degrees of freedom to the choice of alpha, while the cross-bloc relationships remain fully data-driven. The calibration problem reduces to recovering the vector of alpha values that make the model reproduce the target moments which is precisely what the SMM procedure does.

Global Params

Global parameters govern the structure of the game rather than the characteristics of individual players. They cannot be identified through cross-bloc comparisons, so each is anchored to a baseline drawn from published estimates in the climate economics and game theory literature. These baselines are reference points, not fixed values. The global parameters are treated as experimental inputs and varied systematically across the analysis.

Three structural features of the game turn on these parameters. The threshold and curvature of the collective benefit function determine whether coordination emerges gradually or through sharp tipping. The learning rate governs how quickly transition costs decline as adoption spreads. The discount rate captures how strongly time preferences shape the tradeoff between early adjustment costs and future stabilisation benefits. Together they define the strategic terrain on which the per-player asymmetries play out. The sensitivity analysis in the results section stress-tests the findings across the range of plausible values for each.

Special Note on Player Discount Rates

Discount factors follow Nordhaus (2007), reflecting observed political and economic time horizons rather than prescriptive ethics. A game-theoretic framework concerned with strategic behaviour belongs on the descriptive side of this debate. Each bloc's discount factor combines pure time preference with institutional commitment credibility, and because each model period spans five years, delta represents a compounded annual rate rather than a per-period one.

The EU receives $\delta = 0.85$ (3.25% annually), reflecting the legal bindingness of the European Climate Law and over two decades of continuous ETS operation. China is assigned $\delta = 0.80$ (4.6%), where centralised governance supports continuity through the five-year plan system but growth constraints shorten the effective horizon. The US receives $\delta = 0.75$ (5.9%), incorporating a reversal premium for the institutional instability illustrated by withdrawal from and re-entry into the Paris Agreement. The Rest of the World is assigned $\delta = 0.65$ (9.0%), consistent with sovereign borrowing costs in developing economies and compressed fiscal planning horizons (Ameli et al., 2021).

The ordering EU > China > US > RoW holds across all specifications and generates the central strategic tension the model is designed to illuminate: the blocs with greatest patience are not those with greatest climate vulnerability.

Coordination Threshold

The coordination threshold (theta = 0.80) defines the weighted participation level at which stabilisation benefits become substantial. The Paris Agreement's 55% emissions threshold governs entry into force, but entry into force is not effective mitigation. Net-zero pathways in IEA (2021) and IPCC 1.5C scenarios consistently require near-universal engagement among major emitters. The four blocs modelled here account for the substantial majority of global emissions, so a threshold of 0.80 reflects the near-universal requirement those pathways imply. As with other global parameters, tau is varied systematically in the sensitivity analysis to examine how the stringency of the coordination requirement shapes equilibrium outcomes.

Sigmoid Steepness

The sigmoid steepness parameter (η) determines how sharply stabilisation benefits concentrate around the threshold. High values produce tipping-point dynamics, where benefits activate rapidly once θ is crossed. Lower values generate a more gradual accumulation of benefits across adoption levels. The baseline (η = 15) implies a steep transition, with S(W) rising from approximately 0.06 to 0.94 within a ±0.2 band around θ. Since this parameter governs the qualitative shape of payoffs rather than a directly observable quantity, it is treated as an experimental lever.

QRE Rationality

The precision parameter lambda governs how sharply each bloc's strategy responds to payoff differences. Higher values push behaviour toward deterministic best-response; lower values imply greater noise in strategic choice. Experimental estimates typically range from 0.5 to 10 depending on context (McKelvey and Palfrey, 1995; Goeree, Holt, and Palfrey, 2003).

The baseline sets lambda = 1.5 as a common global benchmark, placing it toward the lower end of the experimental range. This reflects the persistent noise in international climate negotiations: domestic political constraints, incomplete information, electoral volatility, and bureaucratic friction all prevent governments from acting as perfectly rational best-responders even when the strategic logic of cooperation is clear.

The model also allows lambda to vary at the bloc level. Bloc-specific precision captures heterogeneity in institutional stability, political coherence, and decision-making capacity that a single global parameter cannot. This makes it possible to examine how differences in political rationality across actors shape the timing and likelihood of global coordination, rather than assuming all blocs navigate strategic uncertainty with equal precision.

Technology Learning Rate

The learning rate ($\gamma = 0.25$) captures cost reductions driven by cumulative adoption through spillovers, supply chain development, and learning-by-doing. Empirical estimates suggest cost declines of 20–25% per doubling of capacity for solar PV and 10–15% for wind (Rubin et al., 2015; IRENA, 2023). The chosen value aligns with the upper end of these estimates, reflecting that "adoption" in the model represents a broad technological transition rather than a single technology.

Damage Escalation

The damage escalation rate ($\kappa = 0.05$) governs how climate damages compounds under delayed action. Over the model's 50-year horizon, this implies an increase of roughly 50% in baseline damages under persistent non-cooperation. This magnitude is consistent with IPCC AR6 estimates suggesting losses on the order of 1–2% of GDP per decade of delayed mitigation.

Layer 2 : Simulated Method of Moments (SMM) Calibration

The cardinal scaling layer is estimated using a simulated method of moments. Four scaling parameters remain free once the ordinal structure is fixed: transition costs, climate damages, political pressure, and coordination benefits. SMM recovers their relative magnitudes by selecting the parameter vector that minimises the weighted squared distance between five model-implied moments and their empirical counterparts.

All other structural parameters are held fixed during estimation, including bloc-specific discount factors, rationality parameters, sigmoid steepness, and damage escalation. This isolates identification onto the four scaling parameters and prevents the cardinal weights from absorbing variation driven by unrelated assumptions.

For any candidate vector, the game is solved under quantal response equilibrium using backward induction, and the resulting strategies are passed through Monte Carlo simulation to generate model-implied moments.

Targeted Moments

Five moments are targeted, chosen because each provides distinct identifying variation for the underlying payoff channels.

Moment 1 targets the EU-US adoption gap in period one. The EU has consistently moved more aggressively than the US despite comparable economic weight (Wurzel et al., 2019), disciplining the balance between transition costs and strategic patience.

Moment 2 targets mean US adoption timing conditional on coordination success. This moment is particularly informative for separating the cost and damage channels: higher transition costs postpone adoption while rising damage accumulation accelerates it. The US exhibits persistent strategic delay relative to the EU despite eventual engagement (Keohane and Victor, 2011; Falkner, 2016).

Moment 3 targets the EU leadership ratio, measuring how much more likely the EU is to adopt in period one relative to its nearest rival. This disciplines the degree to which the fitted economy reproduces Europe as the natural first mover in global climate diplomacy (Backstrand and Elgstrom, 2013).

Moment 4 targets the probability of mid-range stagnation, defined as the coalition reaching meaningful but sub-threshold participation and then stalling. This captures the post-Paris failure mode in which broad nominal participation masked persistent gaps in effective commitment (Falkner, 2016; Rogelj et al., 2016).

Moment 5 targets China's period-one adoption probability at 0.05. Despite leading global renewable capacity additions, China has not submitted a binding near-term decarbonisation commitment and continues to prioritise growth-linked energy expansion. This provides a China-specific identification signal absent from the other four, directly disciplining the cost channel for the model's most consequential bloc (IEA, 2025; WRI, 2025).

Five moments identify four unknown cardinal weights, rendering the system over-identified. This is deliberate. Under exact identification the optimiser always finds a perfect fit regardless of whether the parameters are genuinely identified. The additional moment forces the estimator to find a single parameter vector that fits all five simultaneously, which is only possible if the underlying channels are doing real work.

The moments are not equally scaled. Adoption timing and adoption gaps are measured in periods and ratios respectively, while leadership probabilities are bounded between zero and one. Entering all five moments with equal weight would allow the optimiser to achieve a low objective value by fitting the larger-scale moments precisely while tolerating substantial misfit on the probability-scale moments. To prevent this, moments are weighted to normalise across scales, with the full weight vector reported in the Appendix. The fitted parameter vector is accordingly interpreted as the one that simultaneously reproduces all five observable features of the strategic environment rather than trading off fit across moments of different magnitudes.

Minimisation uses the Nelder-Mead algorithm. Gradient-based methods are unsuitable because the objective is evaluated through Monte Carlo simulation: coordination success rates are computed by simulating discrete adoption paths, producing a noisy objective surface that violates the differentiability assumptions gradient methods require. Nelder-Mead navigates this by evaluating the objective directly without requiring gradient information. To guard against local minima, the procedure is run from multiple random starting points across the parameter space, with the global minimum taken as the estimate. The verification checks reported below confirm the solution is stable across restarts.

The four parameters are not equally precisely identified. Alpha_c and alpha_d are the most tightly identified channels, with adoption gaps and timing moments providing strong direct variation. Alpha_p and alpha_b are identified within broader regions, reflecting the limited direct observability of pressure and benefit scaling in aggregate outcomes. The estimates characterise a convergence region rather than a unique point. The fitted vector is the baseline from which all counterfactual analysis proceeds.

[TABLE DISPLAYING Fitted Baseline vs Targets]

Verification and Stability

Five verification checks are applied following estimation.

The first check restarts the optimiser from the estimated solution. Parameters shift by a maximum of 0.005 and the objective improves marginally, confirming a genuine local minimum.

The second check reruns the model across five random seeds. Coordination success rates range from 72.6% to 73.9% with standard deviation 0.005, confirming Monte Carlo stability.

The third check verifies that estimated parameters preserve all five empirically-grounded ordinal rankings across blocs. All are preserved exactly.

The fourth check perturbs all five target moments by ±10% and records parameter movement. Mean elasticity of 0.20 is well within acceptable bounds, indicating the estimates are stable to small changes in targets.

The fifth check computes the Jacobian condition number as a formal measure of identification strength. The five-moment specification produces a condition number of 199, well within the conventionally acceptable threshold of 500, indicating the moment vector provides genuinely distinct identifying variation rather than near-collinear signals. α_c and α_d carry the strongest identification signals; α_p and α_b are identified within broader regions consistent with the convergence region interpretation adopted throughout. The fifth moment, China's period-one adoption probability, is critical for identification: without it the cost and pressure channels for the model's most consequential bloc are not separately pinned down, and the condition number rises sharply. Its inclusion is what allows the analysis to separately characterise China's strategic delay as a cost problem rather than a patience problem.

[Verification and stability test table]

Experimental Design

What this model is for

The model is not designed as a forecasting tool. It does not attempt to predict whether real-world climate coordination will succeed or fail, and its outputs should not be interpreted as literal probabilities of observed outcomes. Instead, its purpose is mechanism exploration. Starting from the SMM-fitted strategic environment, the objective is to identify which policy channels exert the greatest leverage over coordination dynamics and whether those rankings remain stable once multiple structural assumptions are perturbed.

The experimental design follows directly from this objective. Each experiment begins from the fitted baseline economy, perturbs one or more structural parameters, and records how coordination success, timing, and bloc-level behavior shift in response. The shape of these responses reveals the relative sensitivity of the system to different strategic levers and provides a policy map of which channels can materially reduce coordination risk.

Maringal Parameter Sweeps

The primary analytical tool is the continuous marginal parameter sweep. One parameter is varied continuously across a predefined range while all remaining parameters are held at their SMM-estimated baseline values. At each point, the full dynamic game is re-solved and Monte Carlo simulation records the coordination success rate. The result is a marginal success curve with the parameter value on the horizontal axis and the probability of successful coordination on the vertical axis.

These curves answer a fundamental policy question: how far a given strategic lever must move before the system changes regime. A steep curve implies that small deviations from the fitted baseline generate large changes in coordination probability. A flat curve implies that the channel has limited leverage even under substantial perturbation.

Two summary statistics characterise each curve. The 90% crossing is the percentage deviation from the SMM-fitted baseline required to raise coordination success above 90%, and the 95% crossing is the deviation required to make coordination failure very unlikely. Percentage deviations are computed relative to the SMM-fitted baseline value for each parameter, so a crossing of +10% means the parameter must increase by 10% of its baseline value to reach that frontier. Where parameters have natural bounds, sweep ranges are set to respect them: discount factors are capped at 0.95 to preserve discounting, and cost multipliers are bounded below at zero. Crossings that would require exceeding these bounds are recorded as absent rather than extrapolated.

The channels swept include both bloc-specific and global structural levers: discount factors individually and jointly for the United States and China, cost multipliers, pressure multipliers, the technology learning rate, and the coordination threshold. Together these sweeps provide a detailed elasticity map of the fitted climate coordination economy and identify which strategic levers are capable of shifting the system toward safer equilibrium regions.

Global Sensitivity Analysis

Marginal sweeps isolate one channel at a time. While this cleanly identifies local leverage, it cannot establish whether these rankings remain robust once multiple structural forces move simultaneously. The global sensitivity analysis addresses this by jointly perturbing all major structural parameters around the SMM-fitted baseline.

One thousand parameter vectors are drawn from independent uniform distributions spanning plausible ranges for the key structural inputs. The four cardinal scaling parameters estimated by SMM are held fixed at their fitted values throughout: the GSA is designed to stress-test the structural and behavioural parameters of the game, not to re-estimate the payoff channel weights. All other major inputs are varied, including the four bloc-specific discount factors, bloc-specific cost and pressure multipliers, the technology learning rate, sigmoid steepness, QRE rationality parameters, and damage escalation. Costs and pressure are varied as multiplicative factors applied on top of their SMM-calibrated ordinal values, so the cross-bloc rankings identified in Layer 1 are preserved while the absolute magnitudes shift. Discount factor ranges are set to span empirically plausible variation: δ_{US} (0.65, 0.88), δ_{EU} (0.75, 0.90), δ_{CN} (0.68, 0.88), and δ_{RoW} (0.55, 0.78). Full parameter ranges for all 26 inputs are reported in the Appendix.

For each of the 1,000 draws, the full dynamic game is resolved under quantal response equilibrium and 200 Monte Carlo simulations record coordination outcomes. Spearman rank correlations between each input and each outcome provide a non-parametric measure of global influence that is robust to nonlinearity and does not assume a particular functional relationship between parameters and outcomes. With 1,000 draws, correlations as small as $|\rho| \approx 0.06$ are statistically significant at the 5% level, so the analysis has power to detect even modest structural relationships. A parameter that remains highly ranked in both marginal sweeps and the GSA can be interpreted as a structurally robust policy lever. One that appears powerful in isolation but weak under joint variation suggests strong interaction effects or local nonlinearity around the fitted baseline.

Results

Model Baselines

Before examining parameter perturbations, the SMM-fitted baseline economy is first reported. The model retains the same ordinal cross-bloc structure derived from World Bank and related macroeconomic data, while the cardinal channel weights governing costs, damages, political pressure, and coordination benefits are estimated through the simulated method of moments procedure. Under this fitted strategic environment, coordination succeeds in 71.7% of simulated paths, with a mean coordination timing of 4.20 periods and a mean final coalition weight of 0.69.

The fitted world is best understood through the lens of a chicken game between asymmetric players. Rational actors usually swerve, and the model reflects this: coordination is a likely outcome. The EU moves first at 23.3%, the US follows at 15.7%, China sits back at 8.1%, and RoW is near-inactive at 0.1%. Adoption probabilities decline sharply after period four as the window for successful coordination narrows, with all blocs converging toward low but non-zero probabilities in later periods.

But the interesting question is not whether coordination happens. It is what determines the residual 28% tail risk of catastrophic failure. In a chicken game, the rare outcome where neither player swerves is not just a bad outcome, it is the irreversible one. The purpose of the marginal sweeps and global sensitivity analysis is therefore to identify which strategic levers compress that tail, separating instruments that nudge an already-likely outcome from those that push failure into a genuinely remote event.

[BASELINE FIGURES ADD IN LATEX]

[BASELINE TABLE ADD IN LATEX]

Marginal Sweep Findings

The hierarchy is sharp. A small number of channels can actually compress the dangerous tail of coordination failure to safe levels. Most cannot.

The joint US-China patience channel is the single most powerful lever in the model. Improving both blocs' planning horizons by just 6.75% simultaneously jumps the system across both the 90% and 95% coordination frontiers at the same point. The system does not drift gradually into the safe zone, it tips there once bilateral credibility crosses a threshold. No other channel in the analysis produces this property. Joint declarations, reciprocal ratchet mechanisms, and institutional arrangements that reduce the risk of unilateral reversal are worth more than the sum of their parts precisely because they target this axis directly.

The coordination threshold θ is the strongest single lever, needing only a 6.81% reduction to cross 90% and 11.94% for 95%. Its dominance across every test points to something more interesting than a literal participation requirement. What θ really captures is whether blocs believe meaningful progress is within reach. In a game with threshold dynamics that belief is itself a cause of coordination, not just a consequence of it. Blocs that perceive the goal as achievable adopt earlier, which makes it more achievable, which validates the belief. Policy architectures that make partial progress visible through staged milestones, early coalition wins, and salient intermediate targets are compressing the perceived θ even without changing any underlying physics.

Individual patience channels also reach both frontiers: δ_{US} needs +10.31% for 90% and +21.21% for 95%, δ_{CN} needs +8.52% and +18.75%, δ_{RoW} needs +18.88% and +23.08%. All three reaching 95% confirms patience as a structurally robust instrument rather than a feature of the particular baseline.

Transition costs form a clear second tier. A global cost reduction of 15.45% crosses both frontiers simultaneously. Bloc-specific reductions require larger movements, with CN costs needing 37.27% for 90% and 59.09% for 95%, US costs needing 26.36% and 59.09%, and RoW costs needing 25.45% and 43.64%. Cost reductions matter most as accelerants once the patience and threshold conditions are already shifting rather than as primary tail-risk instruments on their own.

Everything else falls well short. Global pressure requires an extreme 127.27% increase just to cross 90% and 196.36% for 95%. Learning spillovers need 85.44% for 90% and never reach 95%. Damage escalation needs 127.20% for 90% and 161.80% for 95%. These channels amplify coordination once it is already moving rather than compressing the tail independently, a conclusion the GSA confirms below.

One result is excluded from policy interpretation. The δ_{EU} sweep produces an artefact of the flat objective landscape near the baseline rather than a genuine mechanism, inconsistent with the GSA finding that EU patience is essentially uncorrelated with outcomes everywhere. The GSA provides the more reliable picture.

Figures of Key Sweeps:

[ADD IN LATEX]

Tables of Key Sweep params:

[Sweep Table]

GSA Findings

Where the marginal sweeps identify which levers compress the tail in isolation, the GSA tests whether these rankings survive once the entire strategic environment moves simultaneously. The answer is that they do, but with important additions the sweeps alone could not reveal.

The coalition threshold θ dominates both outcome dimensions. It is the strongest correlate of coordination failure (rho = -0.439, $p < 0.001$) and the strongest correlate of coordination delay (rho = +0.565, $p < 0.001$). Across 1,000 joint parameter draws, no single input comes close to matching its influence on whether and when coordination succeeds. The sweep finding that small reductions in θ jump the system across both frontiers simultaneously now has a global counterpart.

The patience axis survives as a first-order driver. δ_US is the strongest positive correlate of coordination success in the entire GSA (rho = +0.338, $p < 0.001$), with δ_CN following at rho = +0.185 ($p < 0.001$). Both materially accelerate coordination timing, confirming the bilateral credibility result from the sweeps is not a local artefact.

The GSA adds a finding the sweeps could not produce: China's domestic transition cost is a distinct and independent bottleneck. Cost_US and Cost_CN are the strongest negative correlates of success after θ, at rho = -0.312 and -0.301 respectively. More strikingly, Cost_CN exhibits the largest negative correlation with China's first-mover probability of any parameter in the entire analysis (rho = -0.540, $p < 0.001$). China is not holding out purely because it is strategically patient. Domestic decarbonisation is genuinely expensive, and patience and cost are jointly constraining Chinese first-mover behaviour through mechanisms that require different policy instruments to address. Bilateral credibility mechanisms target the patience dimension; technology transfer, concessional green finance, and supply chain cooperation target the cost dimension. Both are necessary.

The damage escalation parameter κ produces an unexpectedly strong positive correlation with coordination success (rho = +0.274, $p < 0.001$) despite requiring extreme deviations in the marginal sweeps. This divergence is itself informative: κ operates through interaction effects, amplifying the effectiveness of patience and cost channels once the strategic environment is already close to tipping. Policies that make the costs of delay more immediate are best understood as multipliers on a functioning credibility architecture rather than standalone instruments.

The rationality parameters tell a more nuanced story. λ_US shows meaningful positive correlation with Chinese first-mover probability (rho = +0.149, $p < 0.001$) and accelerates coordination timing (rho = -0.216, $p < 0.001$). A more strategically precise US disciplines Chinese expectations and sharpens the cascade. By contrast, λ_CN is negatively correlated with Chinese first-mover probability (rho = -0.243, $p < 0.001$): a more rational China calculates more precisely that waiting dominates moving first. Structural counterfactual tests confirm this asymmetry survives a full swap of US and China parameters, meaning it reflects a genuine property of how rationality interacts with the free-rider incentive rather than an artefact of China's structural position. Institutional reforms that reduce noise and volatility in US climate policymaking therefore have spillover benefits for Chinese strategic behaviour beyond their direct effect on US adoption.

Pressure channels are confirmed as irrelevant across all three outcome dimensions. All four pressure parameters produce correlations indistinguishable from zero. Pressure is an emergent consequence of coordination rather than a driver of it, and this conclusion holds across both the local sweep and the global joint variation test.

δ_EU remains essentially zero across all outcomes (rho = +0.010, p = 0.759). The EU moves first because of its ordinal position in costs, damages, and patience combined, not because its discount factor sits at any particular level.

The sweep and GSA findings converge on a single diagnosis. The 28% tail risk of catastrophic coordination failure is determined by three things: the perceived achievability of the coordination goal, the bilateral credibility of the US-China axis, and China's domestic transition burden. Everything else operates as a complement to this core rather than a substitute for it.

[GSA FIGURES]

[GSA TABLE]

Functional Form Robustness

Three functional form assumptions are stress-tested: the shape of collective benefits, the activation profile of political pressure, and the cost-learning curve. In each case alternative specifications are substituted while all parameters, including the cardinal scaling weights estimated by SMM, are held at their fitted baseline values without recalibration. Holding α_b fixed when switching benefit specifications mechanically favours the sigmoid, since the scaling parameter was estimated under that functional form. This makes the test conservative: if the policy hierarchy survives despite this disadvantage, the result is stronger than it would be under a recalibrated alternative.

The benefit specification is the most consequential. The baseline sigmoid $b \cdot S(W)$ is tested against a linear specification $b \cdot W$, where every unit of adoption delivers equal marginal returns, and a convex specification $b \cdot W^2$, which returns back-load toward near-universal coordination. These three forms span the space of beliefs about climate benefit aggregation: threshold concentration, uniform accrual, and back-loaded returns. The sigmoid, linear, and convex alternatives produce baseline success rates of 72.6%, 58.1%, and 68.3% respectively, and the magnitudes of key correlations shift across specifications. The policy hierarchy does not. Patience and cost channels dominate pressure and learning channels regardless of which form is assumed. The θ finding is the single exception: its policy relevance depends on benefits concentrating near a participation frontier, which is a question about climate physics rather than strategic structure. A policymaker who rejects the tipping point literature should reach identical conclusions about which levers to prioritise.

The pressure test substitutes the baseline linear specification $p_i \cdot W_t$ with a convex alternative $p_i \cdot W_t^2$, under which pressure remains weak for most of the game and only becomes substantial once the coalition is already large. Under the convex specification all four pressure correlations become statistically indistinguishable from zero. This is not evidence that pressure is suppressed under the linear baseline. It is evidence that convex pressure activates too late to do independent work. By the time coalition weight is large enough to generate meaningful pressure, patience and benefit dynamics have already determined the outcome. Both specifications confirm the GSA finding: pressure emerges from coordination rather than driving it.

The cost-learning curve is tested across three alternatives. The baseline linear form $c_i(W_t) = c_i^0(1 - \gamma W_t)$ implies equal cost reductions per unit of adoption. The convex form $c_i(W_t) = c_i^0(1 - \gamma W_t^2)$ front-loads costs by making early adoption expensive and late adoption cheap. The concave form $c_i(W_t) = c_i^0(1 - \gamma\sqrt{W_t})$ back-loads reductions by delivering the steepest cost falls at low adoption levels. GSA rankings are essentially identical across all three, with the same sign, same magnitude, and same significance for every parameter. The γ correlation remains weak regardless of specification, confirming that learning spillovers are genuinely a minor driver rather than an effect masked by functional form.

[Functional Form Figures Benefit]

[Functional Form Figures Pressure]

[Functional Form Figures Cost]

Equilibrium Uniqueness

Quantal response equilibrium is defined as a fixed point in the mapping from beliefs to logit best-response probabilities. Fixed-point problems of this kind can in principle admit multiple solutions, meaning the model could characterise different equilibria depending on where the iteration is initialised. If multiple fixed points exist, the findings would depend on equilibrium selection rather than the underlying strategic structure, which would undermine the policy conclusions. Two tests are applied to rule this out.

The first initialises the fixed-point iteration from 26 starting points at each of the 150 nodes in the game tree, including the corners of the probability simplex and random interior points. All initialisations converge to identical equilibrium probability vectors, with a maximum deviation of 9.62×10^{-9} across all nodes and starting points. The second test computes the spectral radius of the Jacobian of the logit best-response map at the converged solution. A spectral radius below one confirms the fixed point is a strict contraction, meaning the iteration is guaranteed to converge to a unique solution regardless of initialisation. The maximum spectral radius across all nodes is 0.404, well inside the unit circle, with a median of 0.10. Together these results establish that the equilibrium characterised throughout the analysis is unique: the findings do not depend on equilibrium selection and there is no multiplicity problem to resolve.

{Equilbrium Uniqueness Table]

Policy Implications

The model produces a single clear diagnosis: in a coordination game where rational players usually swerve, the 28% tail risk of catastrophic failure is not randomly distributed across policy levers. It is concentrated along three axes, and compressing it requires targeting all three simultaneously rather than treating them as substitutes.

Prioritise the US-China credibility axis

The joint US-China patience channel is the only lever that jumps the system across both the 90% and 95% tail-risk frontiers at the same point, with a deviation of just +6.75% from baseline. No other intervention achieves this. The GSA confirms δ_{US} as the single strongest positive correlate of coordination success across 1,000 joint parameter draws. The prescription is direct: the highest-return climate investment available is institutional reform that extends the effective planning horizon of the United States and makes that extension credible to China. Legally binding domestic commitments, bipartisan climate frameworks, and reciprocal ratchet mechanisms with China all target this axis. The rationality asymmetry finding adds a further dimension: reducing noise and volatility in US policymaking has spillover benefits for Chinese strategic behaviour beyond the direct effect on US adoption, because a more deliberate US disciplines Chinese expectations in ways that a more deliberate China does not reciprocate. Reforms that stabilise US climate policy are therefore doing double duty.

Address China's cost problem separately

The patience prescription is necessary but not sufficient. $Cost_{CN}$ exhibits the largest negative correlation with China's first-mover probability of any parameter in the analysis ($\rho = -0.544$), and this effect survives when patience is held constant. China is not holding out purely because it is strategically impatient. It is holding out because domestic decarbonisation is genuinely expensive relative to its development position. This is a different problem requiring different instruments: technology transfer agreements, concessional green finance, supply chain cooperation, and grid investment support that directly reduce the domestic burden of early movement. Bilateral credibility mechanisms alone will not solve it. A US-China climate agreement that addresses planning horizons without addressing Chinese transition costs is targeting one bottleneck while leaving the other intact.

Make the goal feel achievable

θ dominates every test in this analysis, and its policy interpretation is not about literally changing a participation requirement. It is about whether blocs perceive meaningful gains from moving before near-universal participation is reached. In a game with threshold dynamics, this perception is a causal input into whether the threshold gets reached. Policy architectures that make partial progress salient, through visible milestones, staged benefit mechanisms, early coalition wins, and credible roadmaps, are compressing the perceived distance to success even without changing the underlying physics. The Paris Agreement's ambition mechanism, the IEA net-zero roadmap, and the proliferation of sectoral coalitions are all attempts to shift this perception. The model suggests they are targeting the right variable.

Use everything else as amplification

Transition cost reductions beyond China, damage urgency mechanisms, carbon border adjustments, and technology learning spillovers all appear in the model as interaction effects rather than standalone levers. They strengthen coordination cascades once the credibility and perception conditions are already shifting but cannot substitute for them. The prescription is to sequence interventions accordingly: establish bilateral US-China credibility and reduce Chinese transition costs first, then use cost, pressure, and urgency instruments to accelerate and lock in the cascade once it begins. Deploying amplifiers before the primary levers are in place is the model's clearest description of what current climate policy is doing wrong.